

1. Administrative & Study Identification

Full Study Title: An Open-label, Randomized, Phase 3 Study to Evaluate Patritumab Deruxtecan Monotherapy versus Treatment of Physician's Choice in Hormone Receptor-positive, HER2-negative Unresectable Locally Advanced or Metastatic Breast Cancer **Public Study Title:** A Clinical Study of Patritumab Deruxtecan to Treat Breast Cancer (MK-1022-016) **Short Title/Acronym:** HERTHENA-Breast04 **Protocol Number:** MK-1022-016 **NCT Number:** NCT07060807 **Sponsor Name:** Merck (Merck Sharp & Dohme LLC) **Investigational Product:** Patritumab deruxtecan (also known as HER3-DXd or MK-1022) **Phase of Development:** Phase 3 **Medical Monitor:** Merck Sharp & Dohme LLC Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate whether patients who receive patritumab deruxtecan live longer overall (Overall Survival [OS]) or without the cancer growing/spreading (Progression-Free Survival [PFS]) compared to those who receive treatment of physician's choice. **Secondary Objectives:** To evaluate Objective Response Rate (ORR), Duration of Response (DOR), change from baseline in the European Organization for Research and Treatment of Cancer (EORTC)-Quality of Life Questionnaire-Core 30 (QLQ-C30) Global Health Status (Item 29) and Quality of Life (Item 30) Combined Score, and safety/incidence of adverse events.

2.2 Background & Rationale To address the urgent need for new therapies in patients with unresectable locally advanced or metastatic hormone receptor-positive (HR+), HER2-negative breast cancer. Despite advancements, this patient population faces an overall poor prognosis after their disease progresses following standard endocrine therapy and CDK4/6 inhibitor treatments. This study evaluates if a HER3-directed antibody-drug conjugate (ADC) can provide a meaningful treatment option.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Trial Status:** Recruiting **Control Type:** Active Comparator (Treatment of physician's choice, which may include paclitaxel, nab-paclitaxel, capecitabine, doxorubicin/liposomal doxorubicin (e.g., LipoDox, Caelyx, Doxil, Adriamycin), or trastuzumab deruxtecan) **Blinding:** Open-label **Randomization:** Randomized (stratified by HER2

expression, treatment intent, HER3 expression, and presence of visceral disease).

3.2 Planned Enrollment & Duration Target **\$N\$:** 1,000 participants
Number of Sites: Global sites across Asia, Europe, North America, and South America
Estimated Study Start: 21-Jul-2025
Estimated Primary Completion: 14-Jul-2033

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age $\geq$ 18 years. 2. Confirmed diagnosis of HR-positive (HR+)/HER2-negative (IHC 0, IHC 1+ or IHC 2+/ISH-) invasive breast carcinoma that is unresectable locally advanced or metastatic. 3. Must have had disease progression or recurrence on prior CDK4/6 inhibitor + endocrine therapy (ET).

4.2 Exclusion Criteria 1. Has breast cancer amenable to treatment with curative intent. 2. Is eligible to receive additional endocrine-based treatment in the advanced setting. 3. Has received prior chemotherapy for unresectable locally advanced/metastatic disease or prior treatment with an anti-HER3 antibody or topoisomerase I inhibitor ADC.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Patritumab deruxtecan 5.6 mg/kg administered on Day 1 of every 3-week cycle.
Route of Administration: Intravenous (IV). **Duration of Treatment:** Cycles repeat every 3 weeks until disease progression or unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Standard palliative and supportive care for symptoms. **Prohibited:** Concurrent systemic anti-cancer therapy, other investigational agents, or chronic systemic steroid therapy.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline Imaging, Tumor Biopsy
Baseline	Day 0	Randomization, First Dose
Interim	Every 3 Weeks	Safety Labs, Efficacy Assessment, Study Drug Administration (IV), Quality of Life assessments
End of Study	End of Treatment	Final Vital Signs, Exit Interview, IP Accountability, Follow-up for Survival

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression-Free Survival (PFS) as assessed by Blinded Independent Central Review (BICR) and Overall Survival (OS). **Measurement Method:** Radiographic imaging per RECIST 1.1 and continuous survival tracking.

7.2 Secondary & Exploratory Endpoints - Objective Response Rate (ORR) - Duration of Response (DOR) - Change from Baseline in the European Organization for Research and Treatment of Cancer (EORTC)-Quality of Life Questionnaire-Core 30 (QLQ-C30) Global Health Status (Item 29) and Quality of Life (Item 30) Combined Score - Incidence of adverse events (AEs) and treatment discontinuations due to AEs

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via physical examinations and labs; strict monitoring for Interstitial Lung Disease (ILD) / pneumonitis. **Serious Adverse Events (SAEs):** Reporting timeline, typically within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee to regularly assess safety profiles.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy outcomes (PFS and OS). Safety Set for all AE/SAE evaluations. **Sample Size Justification:** $N=1,000$ to ensure sufficient power to detect a statistically significant improvement in PFS and OS. **Handling of Missing Data:** Standard censoring methods for time-to-event outcomes (e.g., Kaplan-Meier estimations).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits by the sponsor to assure protocol adherence. **Audit Trail:** Centralized eCRF locking and data clarification forms tracked via standard audit mechanisms.

11. Study Identifiers & Governance

Primary ID: MK-1022-016 **Posting ID:** 355912441502203212 **Registry Number:** NCT07060807 **Sponsor/Lead Organization:** Merck (Merck Sharp & Dohme LLC) **Study Title:** HERTHENA-Breast04 **Official Regulatory Title:** An Open-label, Randomized, Phase 3 Study to Evaluate

12. Clinical Framework (Breast Cancer Specific)

Primary Condition: Breast Neoplasms (HR+/HER2- Unresectable Locally Advanced or Metastatic Breast Cancer) **Diagnosis Threshold:** Confirmed HR+ and HER2- (IHC 0, 1+ or 2+/ISH-) status with documented disease progression following CDK4/6 inhibitor therapy and endocrine therapy **Medical Methodology:** HER3-directed Antibody-Drug Conjugate (ADC) **Optimization Target:** Maximum extension of Progression-Free Survival (PFS) and Overall Survival (OS)

13. Study Procedures & Methodology

Management Pathway: Standard oncology clinical pathway for advanced/metastatic breast cancer **Education Protocol (HCP):** Training protocols on identification and management of Interstitial Lung Disease (ILD)/pneumonitis **Education Protocol (Patient):** Onsite training regarding signs/symptoms of ILD and prompt reporting of adverse events **Quality Audit Mechanism:** Tracking of HCP protocol adherence and patient response monitoring

14. Observation & Follow-Up Schedule

Total Duration: Follow-up until estimated study completion in 2033 **Onsite Visit Cadence:** Every 3 weeks aligned with treatment cycles **Remote Monitoring Frequency:** Between treatment cycles as required **Checklist Review Interval:** Routine imaging assessments per RECIST 1.1 guidelines

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[06-Apr-2026]				
Clinical Operations Lead	[Name]	_____	[06-Apr-2026]			Lead	
Statistician	[Name]	_____	[06-Apr-2026]				